

Chapter 7

Constant-degree Polynomial Partitioning and Incidences in $\mathbb{C}^2$

In this chapter we study incidences with complex algebraic curves in the complex plane $\mathbb{C}^2$. Most of the time we will think of $\mathbb{C}^2$ as $\mathbb{R}^4$, so this chapter is also our first step in studying incidences in $\mathbb{R}^d$ (where $d \geq 3$). To handle such incidence problems we introduce an alternative way of using polynomial partitioning, which we refer to as *constant-degree polynomial partitioning*.¹ In $\mathbb{R}^2$ this technique leads to incidence bounds that are slightly worse than those presented in Chapter 3. However, in dimension $d \geq 3$ it is often significantly simpler to derive incidence bounds by using constant-degree polynomial partitioning.

7.1 Introduction: Incidence issues in $\mathbb{C}^2$ and $\mathbb{R}^d$

Consider a point-curve incidence problem in $\mathbb{C}^2$. That is, we have a set of complex points and curves in $\mathbb{C}^2$ where each curve is defined by a polynomial in $\mathbb{C}[z_1, z_2]$ (and the variables z_1 and z_2 get values in $\mathbb{C}$). It is tempting to try to imitate the proof from Chapter 3 for incidences in $\mathbb{R}^2$ — combinatorially obtain a weak incidence bound, partition $\mathbb{C}^2$ using polynomial partitioning, and then apply the weak bound separately in every cell. Unfortunately, partitioning polynomials do not exist in complex spaces, since removing a variety from $\mathbb{C}^d$ cannot disconnect it into more than one connected component. That is, we cannot split the space into cells using varieties.

One way to overcome the above issue is to think of $\mathbb{C}^2$ as $\mathbb{R}^4$, since partitioning

¹As usual, there is no standard name for this technique. Some papers refer to it as “low degree polynomial partitioning”.

polynomials do exist in $\mathbb{R}^4$. Specifically, we define four real variables x_1, x_2, y_1, y_2 and rewrite the two complex coordinates as $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$. Let the map $\phi : \mathbb{C}^2 \rightarrow \mathbb{R}^4$ be defined as

$$\phi(x_1 + iy_1, x_2 + iy_2) = (x_1, y_1, x_2, y_2).$$

Using ϕ , we move from a point-curve incidence problem in $\mathbb{C}^2$ to an incidence problem with two-dimensional varieties in $\mathbb{R}^4$. Indeed, consider a curve in $\mathbb{C}^2$ that is defined by a polynomial $f \in \mathbb{C}[z_1, z_2]$. Asking f to vanish on a point $(x_1 + iy_1, x_2 + iy_2)$ is equivalent to asking both the real and the imaginary parts of f to vanish on x_1, y_1, x_2, y_2 . That is, $\phi(\mathbf{V}(f)) \subset \mathbb{R}^4$ is a variety defined by two polynomials in $\mathbb{R}[x_1, y_1, x_2, y_2]$ of degree at most $\deg f$. Note that ϕ maintains the point-curve incidences.

We now point out a basic issue that arises when studying incidence problems in dimension $d \geq 3$. For simplicity, we consider one of the simplest cases: incidences between points and planes in $\mathbb{R}^3$. Let $\ell \subset \mathbb{R}^3$ be a line, let $\mathcal{P}$ be a set of m points on ℓ , and let Π be a set of n planes that contain ℓ (e.g., see Figure 7.1). This construction satisfies $I(\mathcal{P}, \Pi) = mn$, implying that the problem is trivial.

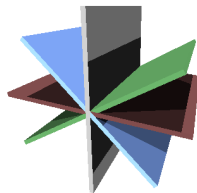


Figure 7.1: By having planes that contain a common line, we can obtain mn point-plane incidences.

To turn the point-plane problem into a non-trivial one, one adds additional restrictions on the points and planes. The most common restriction is probably requiring the incidence graph of $\mathcal{P} \times \Pi$ not to contain a copy of $K_{s,t}$ (for some constants s and t). This problem is interesting (and open), and to obtain some bound for it we try to adapt our proof for the case of point-curve incidences in $\mathbb{R}^2$. Such a restriction arises naturally in many cases. For example, assuming that no k points of $\mathcal{P}$ are collinear implies that the incidence graph contains no $K_{k,2}$. Below we will see that this restriction is also useful when studying incidences in $\mathbb{C}^2$.

Now that we have a non-trivial point-plane incidence problem in $\mathbb{R}^3$, we would like to study it using polynomial partitioning as in Chapter 3. By inspecting the proof of

our weak incidence bound (Lemma 3.4), we notice that it is not only valid for curves in $\mathbb{R}^2$, but rather for any set of varieties in $\mathbb{R}^d$ and also after replacing $\mathbb{R}$ with another field. In particular, this lemma is just a bound on the number of edges in a bipartite graph with no copy of $K_{s,t}$, and does not involve any geometry.

Lemma 7.1. *Let $\mathbb{F}$ be a field and let d be a positive integer. Let $\mathcal{P}$ be a set of m points in $\mathbb{F}^d$ and let $\mathcal{V}$ be a set of n subsets of $\mathbb{F}^d$. If the incidence graph of $\mathcal{P} \times \mathcal{V}$ contains no copy of $K_{s,t}$, then*

$$I(\mathcal{P}, \mathcal{V}) = O_{s,t} \left(mn^{1-\frac{1}{s}} + n \right).$$

The polynomial partitioning theorem also holds in any dimension, and we can use it to partition $\mathcal{P}$. In fact, repeating the analysis for incidences in the cells of the partition as in Chapter 3 is straightforward. On the other hand, the last step of the analysis — handling incidences on the partition itself — becomes significantly more difficult than in the planar case. The partition is a two-dimensional variety, possibly of a large degree. It can contain most of the points of $\mathcal{P}$ and intersect most of the planes of Π in high-degree curves.

When deriving incidence bounds in dimension $d \geq 3$, usually the main difficulty is bounding the number of incidences on the partition. Solymosi and Tao [90] introduced the constant-degree polynomial partitioning technique to overcome this difficulty.

7.2 Curves in higher dimensions

In this brief section we present a generalization of the point-curve incidence bound in $\mathbb{R}^2$ (Theorem 3.3) to higher dimensions. We will rely on this result below, and it is also a good introduction to projections of varieties onto lower dimensional spaces.

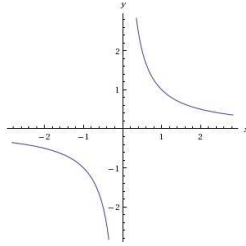


Figure 7.2: The curve $V(xy - 1)$.

Let $U \subset \mathbb{R}^d$ be a variety of degree k and of dimension d' . Let $\pi : \mathbb{R}^d \rightarrow \mathbb{R}^e$ be a standard projection: a linear map that keeps e out of the d coordinates of a point in

$\mathbb{R}^d$. The projection $\pi(U)$ is not necessarily a variety. For example, the projection of $V(xy - 1) \subset \mathbb{R}^2$ onto the x -axis is the set $\{x \in \mathbb{R}^2 : x \neq 0\}$, which is not a variety (see Figure 7.2). However, $\pi(U)$ is *contained* in a variety of dimension at most d' and of degree $O_{k,d}(1)$ (e.g., see [12, Proposition 2.8.6]²). We denote this variety as $\overline{\pi(U)}$.

Lemma 7.2. *Consider an integer $d \geq 2$. Let $\mathcal{P}$ be a set of m points and let Γ be a set of n varieties of dimension at most one and degree at most k , both in $\mathbb{R}^d$. If the incidence graph of $\mathcal{P} \times \Gamma$ contains no copy of $K_{s,t}$, then*

$$I(\mathcal{P}, \Gamma) = O_{s,t,k,d} \left(m^{\frac{s}{2s-1}} n^{\frac{2s-2}{2s-1}} + m + n \right).$$

Proof. We will assume that no two varieties of Γ have a common component. It is not difficult to handle the case where there are common components (see Problem 3.5). We split every reducible one-dimensional variety of Γ into irreducible curves and possibly also one zero-dimensional variety of degree at most k . By Lemma 4.9, every variety of Γ consists of $O_{k,d}(1)$ irreducible components. Thus, after the splitting step we have $|\Gamma| = O_{k,d}(n)$.

We first bound the number of incidences between $\mathcal{P}$ and the zero-dimensional varieties of Γ . By Theorem 4.12, the number of points in such a variety is $O_{d,k}(1)$. Since each zero-dimensional variety participates in $O_{d,k}(1)$ incidences, in total such varieties contribute $O_{d,k}(n)$ incidences.

We remove the zero-dimensional varieties from Γ , so that Γ becomes a set of $O_{k,d}(n)$ irreducible curves. We perform a generic rotation of $\mathbb{R}^d$ around the origin and then project $\mathcal{P}$ and Γ onto the x_1x_2 -plane (that is, we use the projection $\pi(x_1, x_2, \dots, x_d) = (x_1, x_2)$). This process is equivalent to a projection of $\mathcal{P}$ and Γ onto a generic 2-flat of $\mathbb{R}^d$. We set

$$\mathcal{P}' = \{\pi(p) : p \in \mathcal{P}\}, \quad \text{and} \quad \Gamma' = \{\overline{\pi(\gamma)} : \gamma \in \Gamma\}.$$

Projecting the elements of Γ might introduce new intersection points between the curves of Γ' . However, since we first perform a generic rotation, we may assume that the projection does not lead to new incidences. This implies that $I(\mathcal{P}', \Gamma') = I(\mathcal{P}, \Gamma)$ and that the incidence graph of $\mathcal{P}' \times \Gamma'$ does not contain a copy of $K_{s,t}$. Also due to the generic rotation, we may assume that the points of $\mathcal{P}'$ are distinct and that no two curves of Γ' have a common irreducible component. We apply our point-curve incidence result in $\mathbb{R}^2$ (Theorem 3.3) on $\mathcal{P}'$ and Γ' . This yields the assertion of the lemma. $\square$

²I still need to add a proper reference for the degree property.

While Lemma 7.2 is a generalization of Theorem 3.3 to $\mathbb{R}^d$, we can obtain a better bound when $d \geq 3$. For more details see Chapter 11.

7.3 Constant-degree polynomial partitioning

In this section we introduce the constant-degree polynomial partitioning technique. Instead of immediately considering incidences in higher dimensions, we first use this technique to prove a weaker version of the Szemerédi–Trotter theorem. This allows us to see the basic ideas without handling the additional issues that arise in higher dimensions.

Theorem 7.3. *Let $\mathcal{P}$ be a set of m points and let $\mathcal{L}$ be a set of n lines, both in $\mathbb{R}^2$. Then for any $\varepsilon > 0$, we have $I(\mathcal{P}, \mathcal{L}) = O_\varepsilon(m^{2/3+\varepsilon}n^{2/3} + m + n)$.*

A warning readers who are not used to working with $O(\cdot)$ -notation: It is risky to use an $O(\cdot)$ -notation in the statement of the bound that is proved by induction. To see the issue, we now prove the false claim $2^n = O(n)$ by induction on n . For the induction basis, the claim holds for $n \leq 10$ by taking the hidden constant in the $O(\cdot)$ -notation to be sufficiently large. For the induction step, assume that $2^n = O(n)$ for a specific value of n and note that $2^{n+1} = 2 \cdot 2^n = 2 \cdot O(n) = O(n)$. The mistake in the above argument is that the induction step does not really close. While we get an asymptotically correct upper bound for 2^{n+1} , the hidden constant that we get in the $O(\cdot)$ -notation is larger than the constant from the induction hypothesis. That is why in the following proof we require the α variables.

Proof of Theorem 7.3. We prove the theorem by induction on $m + n$. Specifically, we prove by induction that, for any fixed $\varepsilon > 0$, there exist constants α_1, α_2 such that

$$I(\mathcal{P}, \mathcal{L}) \leq \alpha_1 m^{2/3+\varepsilon} n^{2/3} + \alpha_2 (m + n).$$

For the induction basis, the bound holds for small $m + n$ (e.g., for $m + n \leq 100$) by taking α_1 and α_2 to be sufficiently large.

For the induction step, we first recall our weak incidence bound (Lemma 7.1), which implies $I(\mathcal{P}, \mathcal{L}) = O(m\sqrt{n} + n)$. When $m = O(\sqrt{n})$ this implies $I(\mathcal{P}, \mathcal{L}) = O(n)$, which completes the proof. Thus, we may assume that

$$n = O(m^2). \tag{7.1}$$

We take r to be a sufficiently large constant, whose value depends on ε and will be determined below. Let f be an r -partitioning polynomial of $\mathcal{P}$. According to

the polynomial partitioning theorem (Theorem 3.1), f is of degree $O(r)$ and $\mathbf{V}(f)$ partitions $\mathbb{R}^2$ into connected cells, each containing at most m/r^2 points of $\mathcal{P}$. By Warren's theorem (Theorem 3.2), the number of cells is $c = O(r^2)$. The relations between the constants of this proof are³

$$2^\varepsilon \ll r \ll \alpha_2 \ll \alpha_1.$$

Let $\mathcal{L}_0$ denote the subset of lines of $\mathcal{L}$ that are contained in $\mathbf{V}(f)$, and let $\mathcal{P}_0 = \mathcal{P} \cap \mathbf{V}(f)$. Denote the cells of the partition as $C_1, \dots, C_c$. For $j = 1, \dots, c$, put $\mathcal{P}_j = \mathcal{P} \cap C_j$ and let $\mathcal{L}_j$ denote the set of lines of $\mathcal{L}$ that intersect C_j . Note that

$$I(\mathcal{P}, \mathcal{L}) = I(\mathcal{P}_0, \mathcal{L}_0) + I(\mathcal{P}_0, \mathcal{L} \setminus \mathcal{L}_0) + \sum_{j=1}^c I(\mathcal{P}_j, \mathcal{L}_j).$$

For any line $\ell \in \mathcal{L} \setminus \mathcal{L}_0$, by Bézout's theorem (Theorem 2.5) ℓ and $\mathbf{V}(f)$ have $O(r)$ common points. This immediately implies

$$I(\mathcal{P}_0, \mathcal{L} \setminus \mathcal{L}_0) = O(nr). \quad (7.2)$$

Set $m_0 = |\mathcal{P}_0|$ and $m' = m - m_0$; that is, m' is the number of points of $\mathcal{P}$ that are in the cells. Since f is of degree $O(r)$, we get that $\mathbf{V}(f)$ can contain at most $O(r)$ lines. This implies

$$I(\mathcal{P}_0, \mathcal{L}_0) = O(m_0 r). \quad (7.3)$$

It remains to bound $\sum_{j=1}^c I(\mathcal{P}_j, \mathcal{L}_j)$. For $j = 1, \dots, c$, put $m_j = |\mathcal{P}_j|$ and $n_j = |\mathcal{L}_j|$. Note that $m' = \sum_{j=1}^c m_j$, and recall that $m_j \leq m/r^2$ for every $1 \leq j \leq c$. By the induction hypothesis, we have

$$\begin{aligned} \sum_{j=1}^c I(\mathcal{P}_j, \mathcal{L}_j) &\leq \sum_{j=1}^c \left(\alpha_1 m_j^{2/3+\varepsilon} n_j^{2/3} + \alpha_2 (m_j + n_j) \right) \\ &\leq \alpha_1 \left(\frac{m}{r^2} \right)^{2/3+\varepsilon} \sum_{j=1}^c n_j^{2/3} + \alpha_2 \left(m' + \sum_{j=1}^c n_j \right). \end{aligned} \quad (7.4)$$

The above bound of $O(r)$ on the number of intersection points between a line $\ell \in \mathcal{L} \setminus \mathcal{L}_0$ and $\mathbf{V}(f)$ implies that each line enters $O(r)$ cells (a line has to intersect

³By $a \ll b$ we mean that the constant b is larger than some expression depending on the constant a . For example, we might require that $b = \Omega(a^a)$.

$\mathbf{V}(f)$ when moving from one cell to another). This implies $\sum_{j=1}^c n_j = O(nr)$. Combining this with Hölder's inequality (see the “Notation and inequalities” part of the introduction) yields

$$\sum_{j=1}^c n_j^{2/3} \leq \left(\sum_{j=1}^c n_j \right)^{2/3} \left(\sum_{j=1}^c 1 \right)^{1/3} = O\left((nr)^{2/3} \cdot r^{1/3}\right) = O\left(n^{2/3} r^{4/3}\right). \quad (7.5)$$

By combining (7.4) and (7.5), we obtain

$$\sum_{j=1}^c I(\mathcal{P}_j, \mathcal{L}_j) = O\left(\frac{\alpha_1 m^{2/3+\varepsilon} n^{2/3}}{r^{2\varepsilon}} + \alpha_2 nr\right) + \alpha_2 m'.$$

Combining this with (7.2) and (7.3) yields

$$I(\mathcal{P}, \mathcal{L}) = O\left(\frac{\alpha_1 m^{2/3+\varepsilon} n^{2/3}}{r^{2\varepsilon}} + \alpha_2 nr + m_0 r\right) + \alpha_2 m'.$$

By taking α_2 to be sufficiently large with respect to r and the constant in the $O(\cdot)$ -notation, we get

$$\begin{aligned} I(\mathcal{P}, \mathcal{L}) &= O\left(\frac{\alpha_1 m^{2/3+\varepsilon} n^{2/3}}{r^{2\varepsilon}} + \alpha_2 nr\right) + \alpha_2 (m' + m_0) \\ &= O\left(\frac{\alpha_1 m^{2/3+\varepsilon} n^{2/3}}{r^{2\varepsilon}} + \alpha_2 nr\right) + \alpha_2 m. \end{aligned} \quad (7.6)$$

From (7.1) we have $n = n^{2/3} n^{1/3} = O(m^{2/3} n^{2/3})$. By taking α_1 to be sufficiently large with respect to α_2 , r , and the constant in the $O(\cdot)$ -notation in (7.6), we obtain $O(\alpha_2 nr) \leq \frac{\alpha_1}{2} m^{2/3} n^{2/3}$. Similarly, by taking r to be sufficiently large with respect to ε and the constant in the $O(\cdot)$ -notation in (7.6), we may assume that

$$O\left(\frac{\alpha_1 m^{2/3+\varepsilon} n^{2/3}}{r^{2\varepsilon}}\right) \leq \frac{\alpha_1}{2} m^{2/3+\varepsilon} n^{2/3}.$$

Combining this with (7.6) completes the induction step, and thus the proof of the theorem. $\square$

Remarks. (i) Without the extra ε in the exponent of the bound of Theorem 7.3, the induction step would have failed. Specifically, when using the induction hypothesis to sum up the incidences inside of the cells, we would have obtained an expression

that has the correct asymptotic value but with a leading constant that is larger than the one we started with. A similar situation always occurs when using constant-degree polynomial partitioning, and in some sense this is the main disadvantage of the technique.

(ii) Even though we do not apply the weak incidence bound (Lemma 7.1) in every cell as before, the proof still relies on this bound in a different place.

7.4 The Szemerédi–Trotter theorem in $\mathbb{C}^2$

Now that we have a basic understanding of constant-degree polynomial partitioning, we use this technique to handle a more difficult problem — the Szemerédi–Trotter theorem in $\mathbb{C}^2$. That is, we have a point set $\mathcal{P}$ and a set of lines $\mathcal{L}$, both in $\mathbb{C}^2$. One can think of a complex line as the zero set (over the complex numbers) of a linear bivariate polynomial with coefficients in $\mathbb{C}$.

Theorem 7.4. *Let $\mathcal{P}$ be a set of m points and let $\mathcal{L}$ be a set of n lines, both in $\mathbb{C}^2$. Then for any $\varepsilon > 0$, we have $I(\mathcal{P}, \mathcal{L}) = O_\varepsilon(m^{2/3+\varepsilon}n^{2/3} + m + n)$.*

Proof. We move from $\mathbb{C}^2$ to $\mathbb{R}^4$ by using the map ϕ defined in Section 7.1. That is, a point $(x_1 + iy_1, x_2 + iy_2) \in \mathbb{C}^2$ is considered as the point $(x_1, y_1, x_2, y_2) \in \mathbb{R}^4$, and the set $\mathcal{P}$ becomes a set of m points in $\mathbb{R}^4$. Given a line ℓ in $\mathbb{C}^2$, we can write $\ell = \mathbf{V}((a + ia')z_1 + (b + ib')z_2 + (c + ic'))$ for some constants $a, a', b, b', c, c' \in \mathbb{R}$. A point $(x_1 + iy_1, x_2 + iy_2) \in \mathbb{C}^2$ is in ℓ if and only if

$$(a + ia')(x_1 + iy_1) + (b + ib')(x_2 + iy_2) + (c + ic') = 0,$$

or equivalently,

$$bx_2 - b'y_2 + ax_1 - a'y_1 + c = 0 \quad \text{and} \quad b'x_2 + by_2 + a'x_1 + ay_1 + c' = 0.$$

Thus, when moving to $\mathbb{R}^4$ a line ℓ is defined by two linear equations. It is not difficult to verify that each equation defines a distinct hyperplane, and that these two hyperplanes are not parallel. That is, ℓ becomes a 2-flat in $\mathbb{R}^4$. We thus consider $\mathcal{L}$ as a set of n 2-flats in $\mathbb{R}^4$. Note that the set of incidences in $\mathcal{P} \times \mathcal{L}$ does not change when moving from $\mathbb{C}^2$ to $\mathbb{R}^4$. Since two lines in $\mathbb{C}^2$ intersect in at most one point, every two 2-flats of $\mathcal{L}$ intersect in at most one point. That is, the incidence graph of $\mathcal{P} \times \mathcal{L}$ contains no $K_{2,2}$. To complete the proof of the theorem, we derive an upper bound on the number of point-flat incidences in $\mathbb{R}^4$.

An incidence bound in $\mathbb{R}^4$. Imitating the proof of Theorem 7.3, we prove the point-flat incidence bound by induction on $m + n$. Specifically, we prove that for any fixed $\varepsilon > 0$ there exist constants α_1, α_2 such that

$$I(\mathcal{P}, \mathcal{L}) \leq \alpha_1 m^{2/3+\varepsilon} n^{2/3} + \alpha_2(m + n). \quad (7.7)$$

For the induction basis, the bound holds for small $m + n$ by choosing α_1 and α_2 sufficiently large. For the induction step, we recall that the incidence graph contains no copy of $K_{2,2}$. As before, the weak incidence bound (Lemma 7.1) implies $I(\mathcal{P}, \mathcal{L}) = O(m\sqrt{n} + n)$. When $m = O(\sqrt{n})$ this implies $I(\mathcal{P}, \mathcal{L}) = O(n)$, which completes the proof. We may thus assume that

$$n = O(m^2). \quad (7.8)$$

We take r to be a sufficiently large constant, whose value depends on ε and will be determined below. Let f be an r -partitioning polynomial of $\mathcal{P}$. According to the polynomial partitioning theorem, f is of degree $O(r)$ and $\mathbf{V}(f)$ partitions $\mathbb{R}^4$ into connected cells, each containing at most m/r^4 points of $\mathcal{P}$. By Warren's theorem (Theorem 3.2), the number of cells is $c = O(r^4)$. As in the proof of Theorem 7.3, the relations between the constants of this proof are

$$2^\varepsilon \ll r \ll \alpha_2 \ll \alpha_1.$$

Denote the cells of the partition as $C_1, \dots, C_c$. For $j = 1, \dots, c$, put $\mathcal{P}_j = \mathcal{P} \cap C_j$ and let $\mathcal{L}_j$ denote the set of 2-flats of $\mathcal{L}$ that intersect C_j . Let $\mathcal{L}_0$ denote the subset of 2-flats of $\mathcal{L}$ that are contained in $\mathbf{V}(f)$, and let $\mathcal{P}_0 = \mathcal{P} \cap \mathbf{V}(f)$. Notice that

$$I(\mathcal{P}, \mathcal{L}) = I(\mathcal{P}_0, \mathcal{L}_0) + I(\mathcal{P}_0, \mathcal{L} \setminus \mathcal{L}_0) + \sum_{j=1}^c I(\mathcal{P}_j, \mathcal{L}_j).$$

Unlike in the planar case, we cannot rely on Bézout's theorem (Theorem 2.5) to bound the number of cells that are intersected by a 2-flat $h \in \mathcal{L}$. Instead, we notice that every cell of the partition that is intersected by a 2-flat $h \in \mathcal{L}$ corresponds to at least one connected component of $h \setminus \mathbf{V}(f)$. By Theorem 4.11 with $U = h$ and $W = \mathbf{V}(f)$, we get that h intersects $O(r^2)$ cells.

Bounding $\sum_{j=1}^c I(\mathcal{P}_j, \mathcal{L}_j)$. For $j = 1, \dots, c$, put $m_j = |\mathcal{P}_j|$ and $n_j = |\mathcal{L}_j|$. We also set $m' = \sum_{j=1}^c m_j$, and recall that $m_j \leq m/r^4$ for every $1 \leq j \leq c$. By the induction hypothesis, we have

$$\begin{aligned}
\sum_{j=1}^c I(\mathcal{P}_j, \mathcal{L}_j) &\leq \sum_{j=1}^c \left(\alpha_1 m_j^{2/3+\varepsilon} n_j^{2/3} + \alpha_2 (m_j + n_j) \right) \\
&\leq \alpha_1 \left(\frac{m}{r^4} \right)^{2/3+\varepsilon} \sum_{j=1}^c n_j^{2/3} + \alpha_2 \left(m' + \sum_{j=1}^c n_j \right). \tag{7.9}
\end{aligned}$$

The above bound of $O(r^2)$ on the number of cells that are intersected by a 2-flat implies $\sum_{j=1}^c n_j = O(nr^2)$. Combining this with Hölder's inequality (see the “Notation and inequalities” part of the introduction) implies

$$\sum_{j=1}^c n_j^{2/3} = O \left((nr^2)^{2/3} \cdot r^{4/3} \right) = O \left(n^{2/3} r^{8/3} \right). \tag{7.10}$$

By combining (7.9) and (7.10), we obtain

$$\sum_{j=1}^c I(\mathcal{P}_j, \mathcal{L}_j) \leq O \left(\frac{\alpha_1 m^{2/3+\varepsilon} n^{2/3}}{r^{4\varepsilon}} + \alpha_2 nr^2 \right) + \alpha_2 m'.$$

Note that (7.8) implies $n = O(m^{2/3} n^{2/3})$. Thus, by taking α_1 to be sufficiently large with respect to α_2 and r , we have

$$\sum_{j=1}^c I(\mathcal{P}_j, \mathcal{L}_j) \leq O \left(\frac{\alpha_1 m^{2/3+\varepsilon} n^{2/3}}{r^{4\varepsilon}} \right) + \alpha_2 m'.$$

Finally, by taking r to be sufficiently large with respect to ε and the constant of the $O(\cdot)$ -notation, we have

$$\sum_{j=1}^c I(\mathcal{P}_j, \mathcal{L}_j) \leq \frac{\alpha_1}{3} m^{2/3+\varepsilon} n^{2/3} + \alpha_2 m'. \tag{7.11}$$

Bounding $I(\mathcal{P}_0, \mathcal{L} \setminus \mathcal{L}_0)$. If a 2-flat $h \in \mathcal{L}$ is not contained in $\mathbf{V}(f)$, then $\mathbf{V}(f) \cap h$ is at most one-dimensional. Specifically, $\mathbf{V}(f) \cap h$ is a variety of dimension at most one and of degree $O(r)$ (since it is defined by the linear equations of h and by f). We denote the set of these lower-dimensional varieties as $\Gamma = \{h \cap \mathbf{V}(f) : h \in \mathcal{L} \setminus \mathcal{L}_0\}$. We also set $|\mathcal{P}_0| = m_0$. Note that the incidence graph of $\Gamma \times \mathcal{P}_0$ contains no $K_{2,2}$. Thus, Lemma 7.2 implies

$$I(\mathcal{P}_0, \mathcal{L} \setminus \mathcal{L}_0) = I(\mathcal{P}_0, \Gamma) = O_r \left(m_0^{2/3} n^{2/3} + m_0 + n \right).$$

As before, by using (7.8) and taking α_1 and α_2 to be sufficiently large with respect to r and the constant of the $O(\cdot)$ -notation, we have

$$I(\mathcal{P}_0, \mathcal{L} \setminus \mathcal{L}_0) \leq \frac{\alpha_1}{3} m^{2/3} n^{2/3} + \frac{\alpha_2}{2} m_0. \quad (7.12)$$

Bounding $I(\mathcal{P}_0, \mathcal{L}_0)$. Consider a point $p \in \mathcal{P}_0$ such that p is incident to two 2-flats $h, h' \in \mathcal{L}_0$. By performing a translation of $\mathbb{R}^4$ so that p becomes the origin, we can then think of h and h' as vector subspaces. Recall that h and h' intersect only in p , which implies that the vectors from these two spaces span all of $\mathbb{R}^4$. Since both h and h' are contained in $\mathbf{V}(f)$, their tangent planes at p (which are identical to h and h' , respectively) are contained in the tangent hyperplane $T_p \mathbf{V}(f)$. Since two 2-flats cannot span $\mathbb{R}^4$ while also being contained in the same hyperplane, $T_p \mathbf{V}(f)$ is not well defined. We conclude that p is a singular point of $\mathbf{V}(f)$.

The above implies that at most one plane of $\mathcal{L}_0$ can be incident to a point of $\mathcal{P}_0$ that is a regular point of $\mathbf{V}(f)$. Such regular points contribute $O(m_0)$ incidences to $I(\mathcal{P}_0, \mathcal{L}_0)$.

Let V_{sing} be the set singular points of $\mathbf{V}(f)$. By Theorem 4.8, the set V_{sing} is a variety of dimension at most two and of degree $O_r(1)$. Thus, V_{sing} contains $O_r(1)$ 2-flats of $\mathcal{L}_0$ and these yield $O_r(m_0)$ incidences with the points of $\mathcal{P}_0$. The 2-flats of $\mathcal{L}_0$ that are not contained in V_{sing} intersect V_{sing} in varieties that are at most one-dimensional and of degree $O_r(1)$. By Lemma 7.2, the number of incidences that these lower-dimensional varieties contribute is $O_r(m^{2/3}n^{2/3} + m_0 + n)$. By combining the singular and regular cases, we get

$$I(\mathcal{P}_0, \mathcal{L}_0) = O_r(m^{2/3}n^{2/3} + m_0 + n).$$

Once again, by using (7.8) and taking α_1 and α_2 to be sufficiently large with respect to r and the constant of the $O(\cdot)$ -notation, we have

$$I(\mathcal{P}_0, \mathcal{L}_0) \leq \frac{\alpha_1}{3} m^{2/3} n^{2/3} + \frac{\alpha_2}{2} m_0. \quad (7.13)$$

The induction step is obtained by combining (7.11), (7.12), and (7.13), and this in turn completes the proof of the theorem. $\square$

Zahl [108] removed the ε in the bound of Theorem 7.4 by using a more involved analysis. An alternative proof that does not rely on polynomial partitioning and leads to a bound with no ε was obtained by Tóth [101].

Solymosi and Tao proved an incidence result that is more general than the one stated for 2-flats in the proof of Theorem 7.4. The proof of the more general incidence is very similar to the one presented above.

Theorem 7.5 (Solymosi and Tao [90]). *Let $\mathcal{P}$ be a set of m points and let $\mathcal{V}$ be a set of n varieties of degree at most k , both in $\mathbb{R}^d$, such that:*

- *The dimension of every variety of $\mathcal{V}$ is at most $d/2$.*
- *The incidence graph contains no copy of $K_{s,t}$.*
- *There are no incidences between a point $p \in \mathcal{P}$ and a variety $U \in \mathcal{V}$ where p is a singular point of U .*
- *If a point $p \in \mathcal{P}$ is incident to two varieties $U, W \in \mathcal{V}$ and p is a regular point of both U and W , then the tangent spaces $T_p U$ and $T_p W$ intersect only in the origin.*

Then
$$I(\mathcal{P}, \mathcal{V}) = O_{\varepsilon, k, d, s, t} \left(m^{\frac{s}{2s-1} + \varepsilon} n^{\frac{2s-2}{2s-1}} + m + n \right).$$

Note that this result can be seen as a generalization of the planar incidence bound from Theorem 3.3.

7.5 (Optional) Arbitrary curves in $\mathbb{C}^2$

After using constant-degree polynomial partitioning to extend the Szemerédi–Trotter theorem to $\mathbb{C}^2$, we would like to derive a general point-curve incidence bound in $\mathbb{C}^2$. Unfortunately, the proof of Theorem 7.4 does not easily extend even to the case of circles. In that proof we relied on the property that if two 2-flats intersect in a point p , then their tangent planes at p also intersect in a single point. This property holds since if two lines in $\mathbb{C}^2$ intersect in a point p , then their tangent lines at p also intersect in a single point. Hardly any other family of curves in $\mathbb{C}^2$ has this property. For example, two circles in $\mathbb{C}^2$ may intersect in a point p while having the same tangent line at p .

Without the aforementioned tangent intersection property, we are no longer able to use the above argument for bounding the number of incidences with curves that are contained in the partition. To overcome this difficulty, we rely on a more analytic argument from [88]. We begin by defining some basic concepts from Differential Geometry. Since these concepts will not be used anywhere else in the book, we will be somewhat less rigorous than usual.

Intuitively, a set $M \subseteq \mathbb{R}^c$ is a *d -dimensional smooth manifold* if for every point $p \in M$ there exists an open set of M that contains p and “behaves” like a d -flat (specifically, the open set needs to be homeomorphic to an open subset of a d -flat). For our purpose, it suffices to state that any open subset of a d -dimensional variety V that does not contain any singular points of V is a d -dimensional smooth manifold.

For example, any open subset of a plane in $\mathbb{R}^3$ is a 2-dimensional smooth-manifold. An open subset of a circular conical surface is a smooth manifold if and only if it does not contain the apex (see Figure 4.2). For brevity, below we omit the word “smooth” and use the term manifold.

Let M be a d -dimensional manifold. The *tangent bundle* of M is a $2d$ -dimensional manifold defined as

$$TM = \{(p, v) \in M \times (\mathbb{R}^d \setminus \{0\}) : v \in T_p M\}.$$

For $d' < d$, let $E \subset TM$ be a $(d + d')$ -dimensional sub-manifold of TM . We say that E is a d' -dimensional *sub-bundle* of TM if for every point $p \in M$ we have $(\{p\} \times T_p M) \cap E = \{p\} \times V$, where V is a d' -dimensional vector subspace of $T_p M = \mathbb{R}^d$. We denote the subspace V as $E(p) \subset \mathbb{R}^d$.

A *vector field* of a manifold $M \subset \mathbb{R}^c$ is an assignment of a vector of $\mathbb{R}^c$ to each point of M . That is, if X is a vector field on a manifold $M \subset \mathbb{R}^c$, then $X : M \rightarrow \mathbb{R}^d \setminus \{0\}$. As with any function, a vector field X is said to be *smooth* if X is infinitely differentiable. Note that a one-dimensional sub-bundle of a manifold M is a smooth vector field on M . Abusing notation, we sometimes write $X(p) = v$ instead of $X(p) = (p, v) \in TM$. If E is a sub-bundle of TM and $X : M \rightarrow TM$ is a vector field, we say that X *takes values in E* if $X(p) \in E$ for all $p \in M$.

We are only interested in *smooth* vector fields. As with manifolds, we will omit the word smooth for brevity. The following is a variant of the Picard–Lindelöf theorem (e.g., see [58]).

Theorem 7.6. *Let X be a vector field on a manifold M and let $p \in M$. Then for any sufficiently small $\varepsilon > 0$ there exists a unique smooth arc $\alpha : [-\varepsilon, \varepsilon] \rightarrow M$ starting at p whose tangent vectors are in X ; that is, a unique arc α that solves the initial value problem*

$$\alpha(0) = p, \quad \alpha'(t) = X(\alpha(t)) \quad \text{for all } t \in [-\varepsilon, \varepsilon]. \quad (7.14)$$

We will also briefly work with *complex* varieties. The definition of a variety remains identical in this case: Given a set of polynomials $f_1, \dots, f_k \in \mathbb{C}[x_1, \dots, x_d]$, the variety $\mathbf{V}_{\mathbb{C}}(f_1, \dots, f_k)$ is defined as

$$\mathbf{V}_{\mathbb{C}}(f_1, \dots, f_k) = \{(a_1, \dots, a_d) \in \mathbb{C}^d : f_j(a_1, \dots, a_d) = 0 \text{ for all } 1 \leq j \leq k\}.$$

To prevent confusion regarding what field a variety is in, we will sometimes also use the notation $\mathbf{V}_{\mathbb{R}}(f_1, \dots, f_k)$ to refer to a real variety.

Irreducibility, dimension, and singular points of a complex varieties are defined in the same way as for real varieties. For example, a singular point of a variety U is a point where the rank of the Jacobian matrix of U is smaller than the dimension of U (see Section 4.3 for the full definition). For a variety U , recall that we denote the set of singular points of U as U_{sing} , and the set of regular points of U as U_{reg} . Bézout's theorem (Theorem 2.5) and Theorem 4.8 also remain valid for complex varieties.

In the lemma below we have a polynomial $f \in \mathbb{R}[x_1, y_1, x_2, y_2]$ and consider the set $\mathbf{V}_{\mathbb{R}}(f)_{\text{reg}} \setminus \mathbf{V}_{\mathbb{C}}(f)_{\text{sing}}$. This set consists of the regular points of the variety that is defined by f in $\mathbb{R}^4$ that are not singular points of the variety that is defined by f in $\mathbb{C}^4$. For example, let $f = x_1^2 + x_2^2 \in \mathbb{R}[x_1, x_2, x_3]$. Since $\mathbf{V}_{\mathbb{R}}(f)$ is the x_3 -axis, the set $\mathbf{V}_{\mathbb{R}}(f)_{\text{reg}}$ is also the x_3 -axis. Since $\mathbf{V}_{\mathbb{C}}(f)$ is a union of two planes that intersect in the x_3 -axis, the set $\mathbf{V}_{\mathbb{C}}(f)_{\text{sing}}$ is also the x_3 -axis. We conclude that $\mathbf{V}_{\mathbb{R}}(f)_{\text{reg}} \setminus \mathbf{V}_{\mathbb{C}}(f)_{\text{sing}} = \emptyset$.

Recall that the map $\phi : \mathbb{C}^2 \rightarrow \mathbb{R}^4$ is defined by $\phi(x_1 + iy_1, x_2 + iy_2) = (x_1, y_1, x_2, y_2)$. The following lemma will be our main tool for bounding the number of incidences on a constant-sized partitioning polynomial.

Lemma 7.7. *Consider $f \in \mathbb{R}[x_1, y_1, x_2, y_2]$ such that the variety $\mathbf{V}_{\mathbb{R}}(f)$ is three-dimensional. Then for every $p \in \mathbf{V}_{\mathbb{R}}(f)_{\text{reg}} \setminus \mathbf{V}_{\mathbb{C}}(f)_{\text{sing}}$, there is at most one irreducible complex curve $\gamma \subset \mathbb{C}^2$ with $p \in \phi(\gamma)_{\text{reg}}$ and $\phi(\gamma) \subset \mathbf{V}_{\mathbb{R}}(f)$.*

Proof. We set $M = \mathbf{V}_{\mathbb{R}}(f)_{\text{reg}} \setminus \mathbf{V}_{\mathbb{C}}(f)_{\text{sing}}$. By Lemma 4.8, the set $\mathbf{V}_{\mathbb{R}}(f)_{\text{reg}}$ is a three-dimensional manifold and the set $\mathbf{V}_{\mathbb{C}}(f)_{\text{sing}}$ is a variety of dimension at most two. Thus, the set M is a three-dimensional manifold in $\mathbb{R}^4$. The isomorphism ϕ turns multiplication by i in $\mathbb{C}^2$ into the linear transformation

$$J : \mathbb{R}^4 \rightarrow \mathbb{R}^4, \quad J(x_1, y_1, x_2, y_2) = (-y_1, x_1, -y_2, x_2).$$

Notice that for every $u \in \mathbb{R}^4$ we have $J(J(u)) = -u$. Thus, for any linear subspace $U \subset \mathbb{R}^4$ we have $J(J(U)) = U$. Since J corresponds to multiplication by i in $\mathbb{C}^2$, a linear subspace $U \subseteq \mathbb{R}^4$ is J -invariant if and only if $U = \phi(U')$ for some complex subspace $U' \subseteq \mathbb{C}^2$. In particular, all J -invariant subspaces are even dimensional.

For every point $p \in M$ we define the linear subspace $E_p = T_p M \cap J^{-1}(T_p M)$. Intuitively, E_p is the largest linear subspace of $T_p M$ that is invariant under J . Since the linear subspace $T_p M$ is three-dimensional, it cannot be J -invariant. This implies that $J^{-1}(T_p M)$ is a different three-dimensional subspace, and thus E_p is a two-dimensional linear subspace. As p varies, the union of the $p \times E_p$ forms a two-dimensional subbundle E of the tangent bundle TM .

Fix a point $p \in M$, and let X be a vector field such that X is defined in an open neighbourhood $U \subseteq M$ of p , the restriction of X to U takes values in E , and

$X(p) \neq 0$. Note that if such a vector field does not exist, then no irreducible curve $\gamma \subset \mathbb{C}^2$ satisfies $p \in \phi(\gamma)_{\text{reg}}$ and $\phi(\gamma) \subset \mathbf{V}_{\mathbb{R}}(f)$. We thus assume that X exists. By Theorem 7.6 there is a unique arc $\alpha : [-\varepsilon, \varepsilon] \rightarrow \gamma_{\text{reg}}$ that solves (7.14) (for a sufficiently small $\varepsilon > 0$). Two distinct irreducible curves intersect in a zero-dimensional variety, and by Theorem 4.12 this variety consists of finitely many points. Thus, there could be at most one irreducible one-dimensional variety that contains γ . We denote this curve as α^* (if no such curve exists then we are done).

Consider an irreducible complex curve $\gamma \subset \mathbb{C}^2$ that satisfies $\phi(\gamma) \subset \mathbf{V}_{\mathbb{R}}(f)$ and a point $p \in \phi(\gamma)_{\text{reg}}$. For a point $q \in \gamma$, note that if $\phi(q) \in \phi(\gamma)_{\text{reg}}$ then the tangent space $T_q\gamma$ is a complex line in $\mathbb{C}^2$. Since $T_{\phi(q)}\phi(\gamma) = \phi(T_q\gamma)$, the tangent space $T_{\phi(q)}\phi(\gamma)$ is a J -invariant 2-flat in $\mathbb{R}^4$ that is contained in T_qM . This implies that $T_{\phi(q)}\phi(\gamma) = E_q$, so $X(q)$ is tangent to $\phi(\gamma)$. By applying Theorem 7.6 to the manifold $\phi(\gamma)_{\text{reg}}$, and the restriction of X to $\phi(\gamma)_{\text{reg}}$, we obtain an arc $\beta : [-\varepsilon', \varepsilon'] \rightarrow \phi(\gamma)_{\text{reg}}$ that solves the same equation (7.14) (for a sufficiently small $0 < \varepsilon' \leq \varepsilon$). The uniqueness of the solution implies that α^* and β must have a common sub-arc around p .

Assume for contradiction that there exists another irreducible complex curve $\gamma' \subset \mathbb{C}^2$ that satisfies $\phi(\gamma') \subset \mathbf{V}_{\mathbb{R}}(f)$ and $p \in \phi(\gamma')_{\text{reg}}$. By repeating the argument from the previous paragraph, we get another arc $\beta' : [-\varepsilon'', \varepsilon''] \rightarrow \phi(\gamma')_{\text{reg}}$ such that α^* and β' have a common sub-arc α' around p . This in turn implies that β and β' have a common sub-arc around p . We got a contradiction to Bézout's theorem, which states that γ and γ' have a finite intersection. That is, at most one curve γ satisfies the above properties. $\square$

We are finally ready to study incidences with arbitrary curves in $\mathbb{C}^2$.

Theorem 7.8. *Let $\mathcal{P}$ be a set of m points and let Γ be a set of n distinct irreducible algebraic curves of degree at most k , both in $\mathbb{C}^2$. If the incidence graph of $\mathcal{P} \times \Gamma$ contains no copy of $K_{s,t}$, then for any $\varepsilon > 0$ we have*

$$I(\mathcal{P}, \Gamma) = O_{s,t,k,\varepsilon} \left(m^{\frac{s}{2s-1} + \varepsilon} n^{\frac{2s-2}{2s-1}} + m + n \right).$$

Proof. The first half of the proof is very similar to the proof of Theorem 7.3. A reader who wishes to avoid reading this repetition may like to skip to the part titled “Incidences on the partition”.

From Lemma 7.1 we get the weak bound $I(\mathcal{P}, \Gamma) = O_{s,t} \left(mn^{1-\frac{1}{s}} + n \right)$. When $m = O(n^{1/s})$, this implies the bound $I(\mathcal{P}, \Gamma) = O(n)$. We may thus assume that

$$n = O(m^s). \tag{7.15}$$

We will prove by induction on $m + n$ that

$$I(\mathcal{P}, \Gamma) \leq \alpha_1 m^{\frac{s}{2s-1} + \varepsilon} n^{\frac{2s-2}{2s-1}} + \alpha_2(m + n),$$

where α_1, α_2 are sufficiently large constants. The base case where $m + n$ is small can be handled by choosing sufficiently large values of α_1 and α_2 .

We set $\mathcal{P}^* = \{\phi(p) : p \in \mathcal{P}\}$ and $\Gamma^* = \{\phi(\gamma) : \gamma \in \Gamma\}$. Note that $\mathcal{P}^*$ is a set of m points in $\mathbb{R}^4$ and that Γ^* is a set of n two-dimensional varieties of degree $O_k(1)$ in $\mathbb{R}^4$. As in the proof of Theorem 7.3, we will bound $I(\mathcal{P}^*, \Gamma^*)$. Since $\phi : \mathbb{C}^2 \rightarrow \mathbb{R}^4$ is a bijection, we indeed have $I(\mathcal{P}, \Gamma) = I(\mathcal{P}^*, \Gamma^*)$.

Partitioning $\mathbb{R}^4$. Let f be an r -partitioning polynomial of $\mathcal{P}^*$, for a sufficiently large constant r . According to the polynomial partitioning theorem (Theorem 4.10), f is of degree $O(r)$ and $\mathbf{V}(f)$ partitions $\mathbb{R}^4$ into connected cells, each containing at most m/r^4 points of $\mathcal{P}^*$. By Warren's theorem (Theorem 3.2), the number of cells is $c = O(r^4)$. The asymptotic relations between the various constants in the proof are

$$2^{1/\varepsilon} \ll r \ll \alpha_2 \ll \alpha_1.$$

Let $C_1, \dots, C_c$ be the cells of the partition. Let $\mathcal{V}_j$ be the set of varieties from Γ^* that intersect the interior of C_j and let $\mathcal{P}_j$ be the set of points $p \in \mathcal{P}^*$ such that $p^* \in C_j$. Let $m_j = |\mathcal{P}_j|$, $m' = \sum_{j=1}^c m_j$, and $n_j = |\mathcal{V}_j|$. By Theorem 4.10, $m_j = O(m/r^4)$ for every $1 \leq j \leq c$. Notice that

$$I(\mathcal{P}^*, \Gamma^*) = I(\mathcal{P}_0, \Gamma_0) + I(\mathcal{P}_0, \Gamma^* \setminus \Gamma_0) + \sum_{j=1}^c I(\mathcal{P}_j, \Gamma_j).$$

For every $U \in \Gamma^*$, applying Theorem 4.11 with $W = \mathbf{V}(f)$ implies that U intersects $O_k(r^2)$ cells of $\mathbb{R}^4 \setminus \mathbf{V}(f)$. Therefore, $\sum_{j=1}^c n_j = O_k(nr^2)$. Combining this with Hölder's inequality implies

$$\sum_{j=1}^c n_j^{\frac{2s-2}{2s-1}} = O_k \left((nr^2)^{\frac{2s-2}{2s-1}} c^{\frac{1}{2s-1}} \right) = O_k \left(n^{\frac{2s-2}{2s-1}} r^{\frac{4s}{2s-1}} \right).$$

By the induction hypothesis, we have

$$\begin{aligned} \sum_{j=1}^c I(\mathcal{P}_j, \Gamma_j) &\leq \sum_{j=1}^c \left(\alpha_1 m_j^{\frac{s}{2s-1} + \varepsilon} n_j^{\frac{2s-2}{2s-1}} + \alpha_2 (m_j + n_j) \right) \\ &\leq O_k \left(\alpha_1 m^{\frac{s}{2s-1} + \varepsilon} r^{-\frac{4s}{2s-1} - 4\varepsilon} \sum_{j=1}^c n_j^{\frac{2s-2}{2s-1}} \right) + \sum_{j=1}^c \alpha_2 (m_j + n_j) \\ &\leq O_k \left(\alpha_1 r^{-\varepsilon} m^{\frac{s}{2s-1} + \varepsilon} n^{\frac{2s-2}{2s-1}} \right) + \alpha_2 (m' + O_k(nr^2)). \end{aligned}$$

By (7.15), we have $n^{\frac{1}{2s-1}} = O\left(m^{\frac{s}{2s-1}}\right)$, which in turn implies $n = O\left(m^{\frac{s}{2s-1}} n^{\frac{2s-2}{2s-1}}\right)$. Thus, when α_1 is sufficiently large with respect to r and α_2 , we have

$$\sum_{j=1}^c I(\mathcal{P}_j, \Gamma_j) = O_k \left(\alpha_1 r^{-\varepsilon} m^{\frac{s}{2s-1} + \varepsilon} n^{\frac{s}{2s-1}} \right) + \alpha_2 m'.$$

By taking r to be sufficiently large with respect to ε, k , and the implicit constant in the O -notation, we have

$$\sum_{j=1}^c I(\mathcal{P}_j, \Gamma_j) \leq \frac{\alpha_1}{2} m^{\frac{s}{2s-1} + \varepsilon} n^{\frac{2s-2}{2s-1}} + \alpha_2 m'. \quad (7.16)$$

Incidences on the partition. It remains to bound incidences with points that are on the partitioning hypersurface $\mathbf{V}(f)$. Let $\mathcal{P}_0 = \mathcal{P}^* \cap \mathbf{V}(f)$, let $m_0 = |\mathcal{P}_0| = m - m'$, and let $\Gamma_0 = \{U \in \Gamma^* : U \subset \mathbf{V}(f)\}$. By Lemma 7.7, every point of $(\mathcal{P}_0 \cap \mathbf{V}_{\mathbb{R}}(f)_{\text{reg}}) \setminus \mathbf{V}_{\mathbb{C}}(f)_{\text{sing}}$ is a regular point of at most one variety of Γ_0 . That is, these points form at most m_0 incidences with regular points of varieties of Γ_0 . It remains to bound the number of incidences $(p, U) \in \mathcal{P}_0 \times \Gamma^*$ that satisfy one of the following:

- The point p is in $\mathbf{V}_{\mathbb{R}}(f)_{\text{sing}}$ or in $\mathbf{V}_{\mathbb{C}}(f)_{\text{sing}}$.
- The point p is a singular point of U .
- The variety U is not in Γ_0 .

Let $S = \mathbf{V}_{\mathbb{R}}(f)_{\text{sing}} \cup (\mathbf{V}_{\mathbb{C}}(f)_{\text{sing}} \cap \mathbb{R}^4)$. We first consider the case where U is contained in S . Since S is a two-dimensional variety of degree $O_{k,r}(1)$, it contains $O_{k,r}(1)$ varieties of Γ^* . Thus, the total number of incidences with the varieties of Γ^* that are in S is $O_{k,r}(m_0)$.

Next, we set

$$\Gamma_0^* = \{U \cap S : U \in \Gamma_0 \text{ and } U \not\subseteq S\} \cup \{U \cap \mathbf{V}(f) : U \in \Gamma^* \setminus \Gamma_0\} \cup \{U_{\text{sing}} : U \in \Gamma^*\}.$$

Note that Γ_0^* is a set of $O(n)$ varieties in $\mathbb{R}^4$, each of dimension at most one and of degree $O_{r,k}(1)$. This is straightforward for the first two parts in the definition of Γ_0^* , and is implied by Theorem 4.8 for the third part. By Lemma 7.2 we obtain

$$I(\mathcal{P}_0, \Gamma_0^*) = O_{s,t,k,r} \left(m_0^{\frac{s}{2s-1}} n^{\frac{2s-2}{2s-1}} + m_0 + n \right).$$

Combining the three cases above gives

$$I(\mathcal{P}_0, \Gamma^*) = O_{s,t,k,r} \left(m_0^{\frac{s}{2s-1}} n^{\frac{2s-2}{2s-1}} + m_0 + n \right).$$

Taking α_1, α_2 to be sufficiently large with respect to s, t, k, r , and the constant of the O -notation, we have

$$I(\mathcal{P}_0, \Gamma^*) \leq \frac{\alpha_1}{2} m^{k/(2k-1)} n^{(2k-2)/(2k-1)} + \alpha_2(m_0 + n).$$

Combining this with (7.16) completes the proof of the induction step. $\square$

7.6 Exercises

Problem 7.1. The following result is from [45].

Theorem. *Let $\mathcal{P}$ be a set of n points in $\mathbb{R}^2$. Then at most $n^2/6 - n/2 + 1$ lines contain three points of $\mathcal{P}$.*

Use this theorem to prove that the same result holds for any set of n points in $\mathbb{R}^d$ (for every integer $d \geq 3$).

7.7 Open problems

In this chapter we saw how constant-degree polynomial partitioning makes it easier to study incidence problems in dimension $d \geq 3$. In Chapter 11 we will see more advanced uses of this technique for incidences in $\mathbb{R}^d$. This technique has one obvious drawback — it adds an extra ε to the exponent of the incidence bound. As already mentioned, Zahl [108] removed the ε in the special case that arises in the proof of Theorem 7.4. In Section 11.5 we will mention some other cases in which this ε was removed.

Open Problem 7.1. Remove the ε from the exponent in the bound of Theorem 7.8. More generally, develop a technique for removing the extra ε from the exponent whenever constant-degree polynomial partitioning is used.

We saw in this chapter how to derive point-curve incidence bounds in the complex plane $\mathbb{C}^2$. Unfortunately, so far it is not known how to extend our techniques to higher dimensional complex spaces. Elekes and Szabó [33] derived a bound for arbitrary point-variety incidences in $\mathbb{C}^d$. Their analysis relies on older incidence tools, so their bounds are weaker than the bounds obtained in $\mathbb{R}^d$ using polynomial partitioning (see Chapter 11). Dvir and Gopi [28] and Zahl [109] studied a specific point-line incidence problem in $\mathbb{C}^d$, for any $d \geq 3$. However, at the moment we do not have a reasonable bound for what may be considered as the main point-line incidence problem in $\mathbb{C}^3$.

Open Problem 7.2. For every $d \geq 3$, extend the known point-variety incidence bounds in $\mathbb{R}^d$ to $\mathbb{C}^d$.